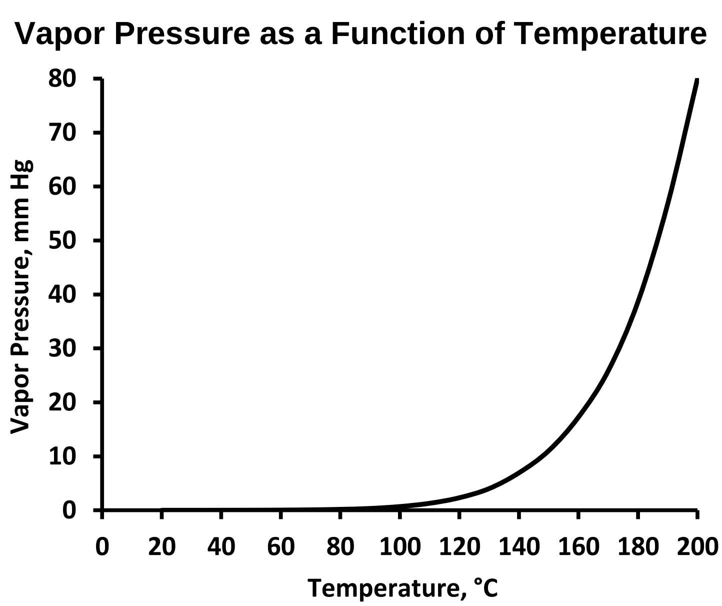
TECHNICAL BULLETIN

DIXIE CHEMICAL
10601 Bay Area Blvd.
Pasadena, TX 77507
Tel: (281) 474-3271
Fax: (281) 291-3384
E-Mail: info@dixiechemical.com

PRE-CATALYZED METHYLHEXAHYDROPHTHALIC ANHYDRIDE (MHHPA 301)

Typical Properties

Property	Value	Test Method
Molecular Weight (average)	170	
Specific gravity @ 25°C, g/cc	1.20	D-102 (ASTM D 4052)
Appearance	clear liquid	D-104 (Visual)
Viscosity (Brookfield at 25 °C), cP	40 - 100	D-150
Color, Pt-Co	< 50	D-103 (ASTM D 1209)



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MHHPA 301

Applications

MHHPA 301 is a low color, low viscosity, pre-catalyzed epoxy curing agent with a proprietary catalyst that provides the following benefits:

- **Easy Handling:** Low viscosity liquid gives low viscosity mixes with various epoxy resins

- **Convenience:** Eliminates the need to add small amounts of catalyst

- **Consistency:** Prevents potential batch-to-batch differences caused by errors in measuring and properly dispersing small amounts of catalyst

- **Low reactivity at room temperature:** Provides long working times and pot lives for mixes with epoxy resins

- **High reactivity at cure temperature:** Gives excellent cure response at cure temperatures, resulting in high crosslinking and relatively high Tg.

- **Low color and low color development:** Low residual unsaturation gives maximum resistance to yellowing, making it desirable where premium optical and electrical performance are needed in epoxy casting and encapsulation applications.

For more information on the use of anhydrides like MHHPA 301 as epoxy curing agents, please consult the Technical Bulletin, FORMULATING ANHYDRIDE-CURED EPOXY SYSTEMS, available from Dixie Chemical Company.

Formulation Example

Most applications use between 80-90 parts by weight MHHPA 301 with 100 parts bisphenol A liquid epoxy resins. MHHPA 301 was formulated with a variety of epoxy resins. Viscosity was measured at 25°C, and gel time was measured at 150°C. Samples were cured for 1 hour at 120°C, followed by a post-cure of 1 hour at 220°C, and glass transition temperatures were measured using a differential scanning calorimeter. The following table summarizes these results:

Epoxy type	MHHPA 301, phr	Viscosity at 25 °C, cP	Gel Time [1], min	Tg [2], °C
Standard BPA Liquid Epoxy	89	1210	19	141
Low Viscosity BPA Liquid	92	845	17	142
Cycloaliphatic Epoxy	122	139	24	206
Epoxy Phenol Novolac	96	968	17	134
Epoxy BPA Novolac	87	3630	17	150
BPF Liquid Epoxy	96	488	15	138

1. Shyodu Hot Pot (100g at 100°C)
2. Cured 1 hr at 120°C and 1 hr at 220°C. Analyzed in DSC: 40°C/min to 250°C

Handling & Storage

MHHPA 301 will react with water to form methylhexahydrophthalic acid. This is normally undesirable, so MHHPA should be stored in such a way that it is carefully protected from moisture contamination.

For more details on the design of bulk storage for MHHPA 301, consult the Dixie Chemical Company brochure "Epoxy Curing Agent Storage Requirements."

Health Hazards

Read and understand the relevant Safety Data Sheets (SDS) for all products before use. These anhydrides are primary skin and eye irritants. Avoid contact with skin, eyes, and clothing. Use only with adequate ventilation. In case of contact, follow the procedures outlined in the SDS. Generally, these procedures include immediately flushing the affected skin or eyes with copious amounts of water for at least 15 minutes. In the case of eye contact, get medical attention. Wash contaminated clothing before reuse.

Follow the recommendations in the SDS for personal protective equipment when handling these materials. At a minimum, these procedures typically include protective chemical goggles, impenetrable gloves, and measures to avoid breathing chemical vapors.

Availability

MHHPA 301 is available from Dixie Chemical Company, Inc.'s plant in Pasadena, Texas. Contact your Dixie Chemical Company representative for details.